

Supplementary information for ChinaTherm30

Supplementary Figures

Six macroregions used for stratification and reporting

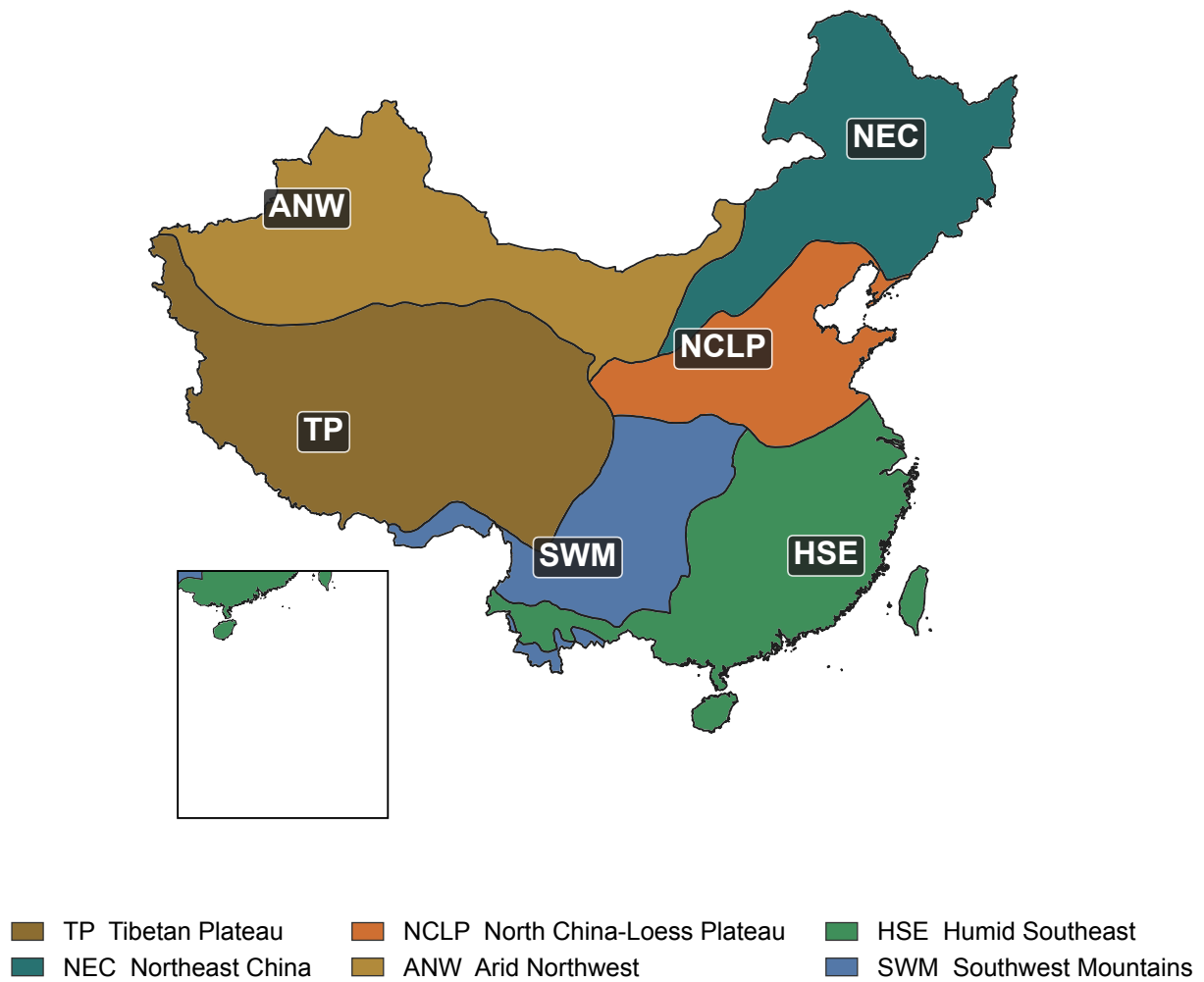


图 S1: Six macroregions used for stratified sampling and regional reporting. The boundaries aggregate 46 ecological regions from the China eco-geographical regionalization: TP, Tibetan Plateau; NEC, Northeast China; NCLP, North China-Loess Plateau; ANW, Arid Northwest; HSE, Humid Southeast; and SWM, Southwest Mountains. These abbreviations are used consistently in the main text and figures.

Spatial support of the 2000-2024 Landsat validation sample

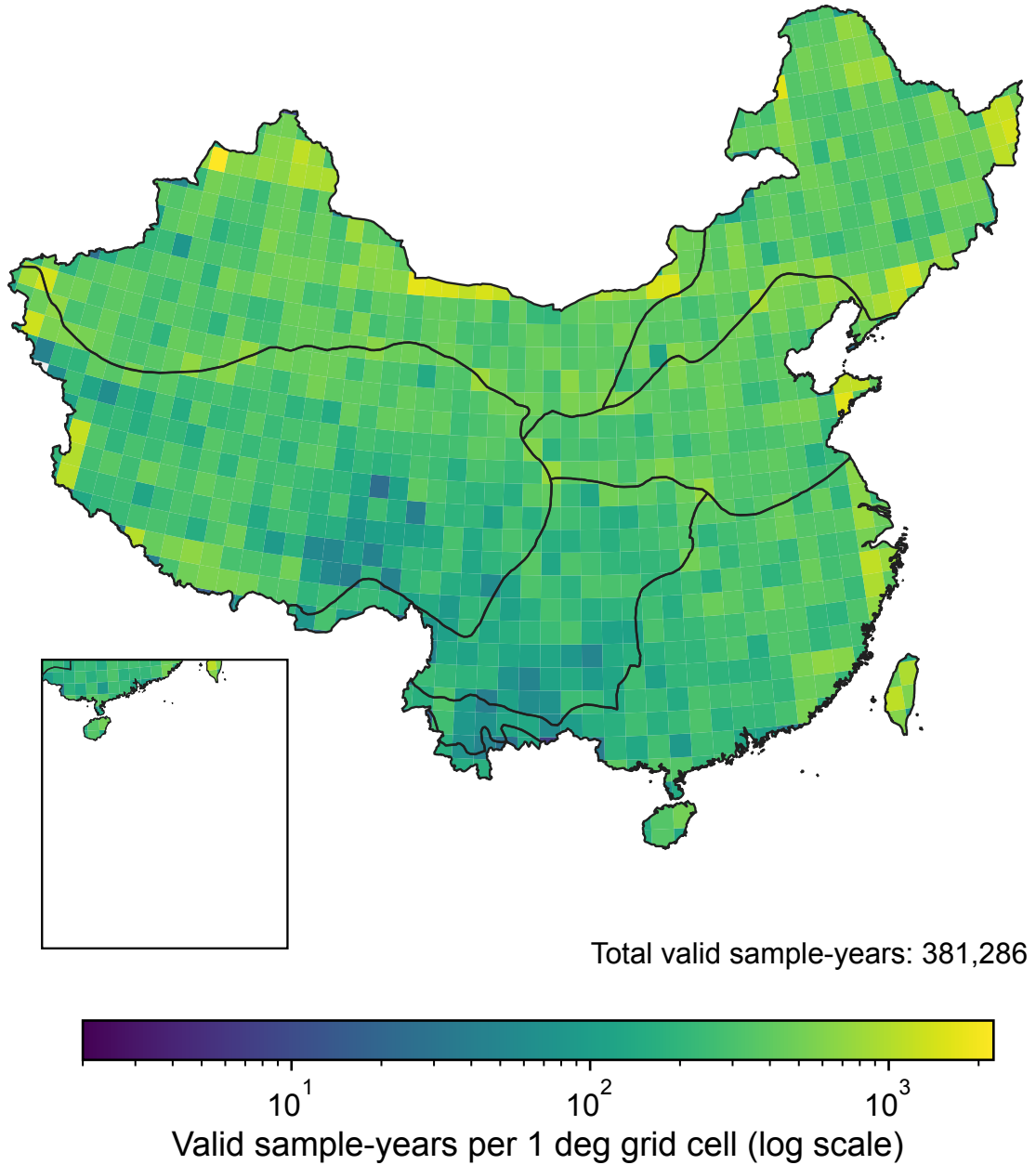


图 S2: Spatial support of the nationwide Landsat Collection 2 surface-temperature validation. Valid samples from 2000–2024 are aggregated to 1° grid cells to avoid overplotting. A sample-year requires strict cloud, shadow, snow and radiometric-saturation screening, at least three valid Landsat observations, available 3×3 ChinaTherm30 and RF-LST baseline means, and a water fraction no greater than 10%. Macroregion boundaries follow Figure S1.

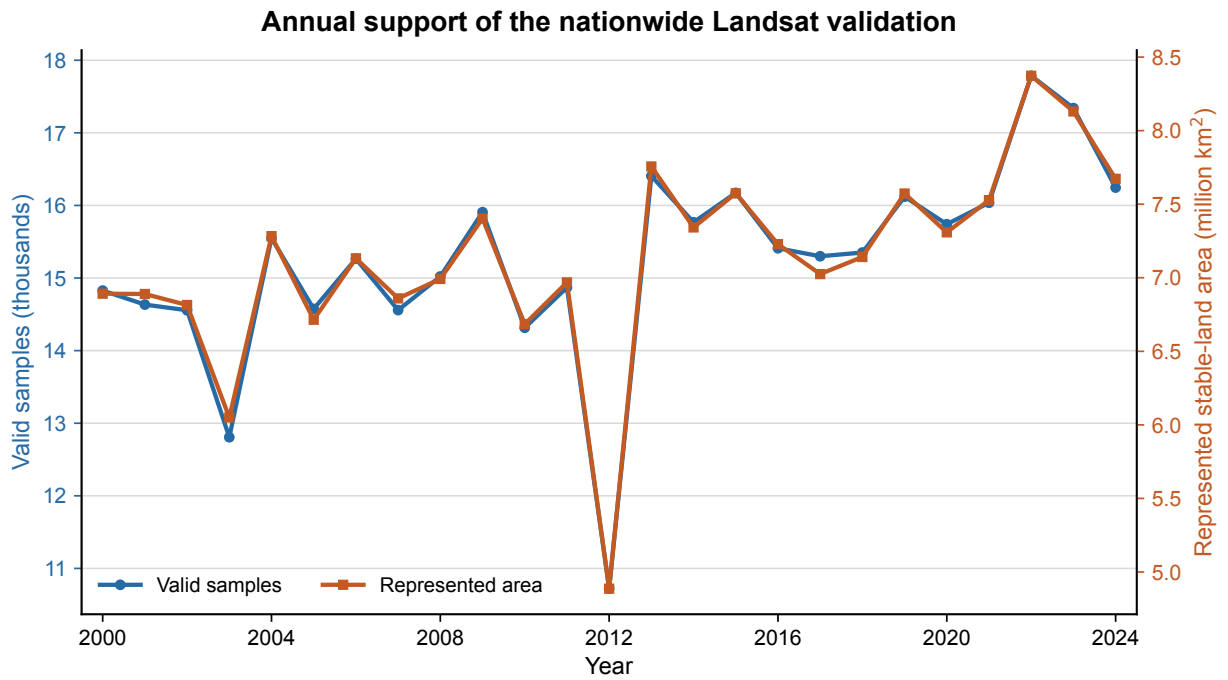


图 S3: Annual support of the nationwide Landsat validation. Valid sample counts and represented stable-land area use the same eligibility and analysis weights as the area-weighted validation metrics in the main text.

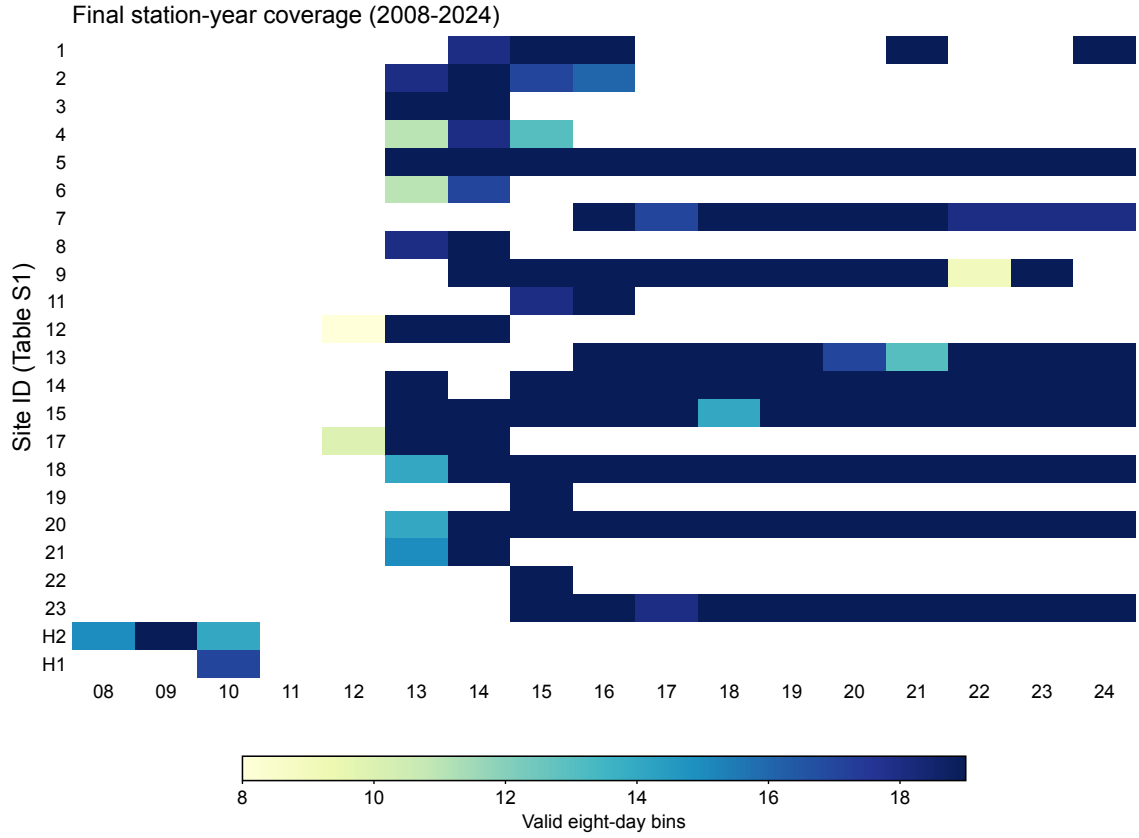


图 S4: Temporal coverage of the final ground-validation set. Colours indicate the number of valid MOD11A2-aligned eight-day bins contributing to each station-year annual IRT composite. The final set contains 131 station-years across all 23 sites and 16 calendar years. Site IDs follow Table S1.

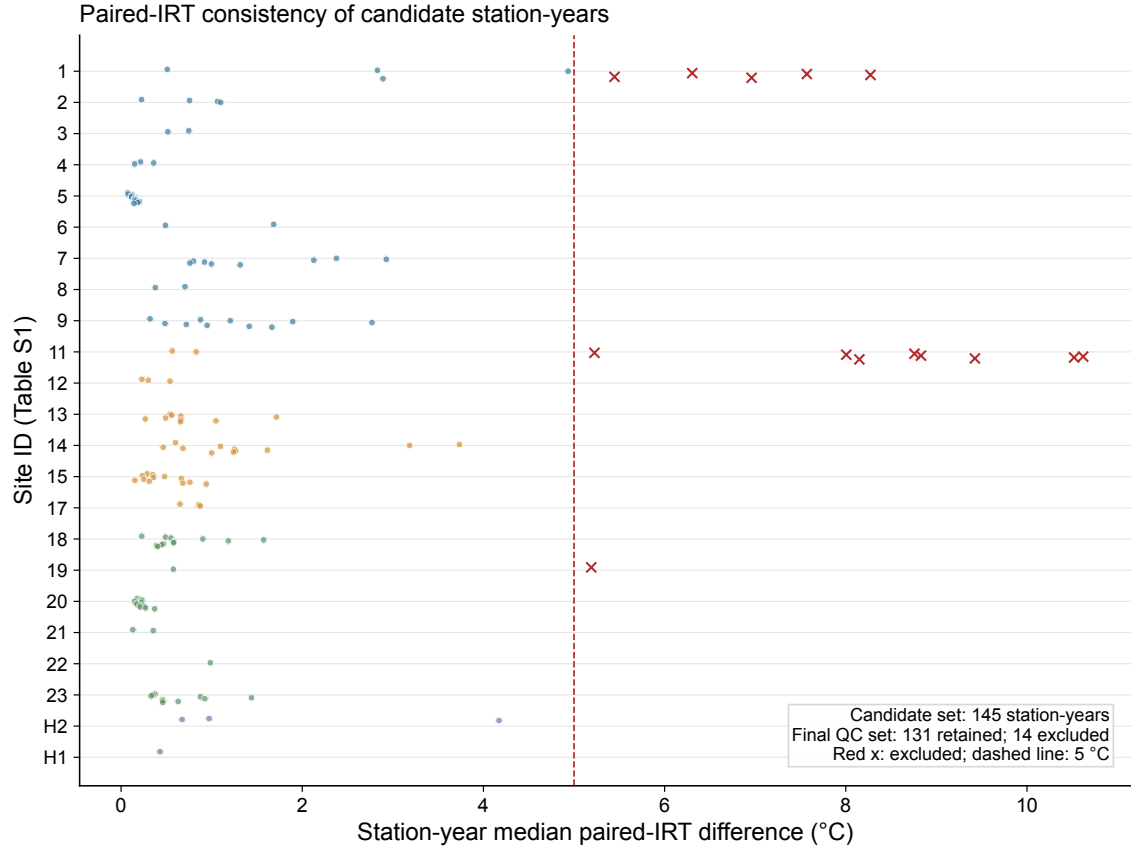


图 S5: Paired-IRT consistency within the 145 candidate station-years. A paired-sensor diagnostic was available for 143 records. Fourteen station-years with a median daily paired-IRT absolute difference greater than 5 °C were excluded from the final ground-reference set; the remaining 131 station-years retained all 23 sites. The criterion uses only ground-sensor measurements.

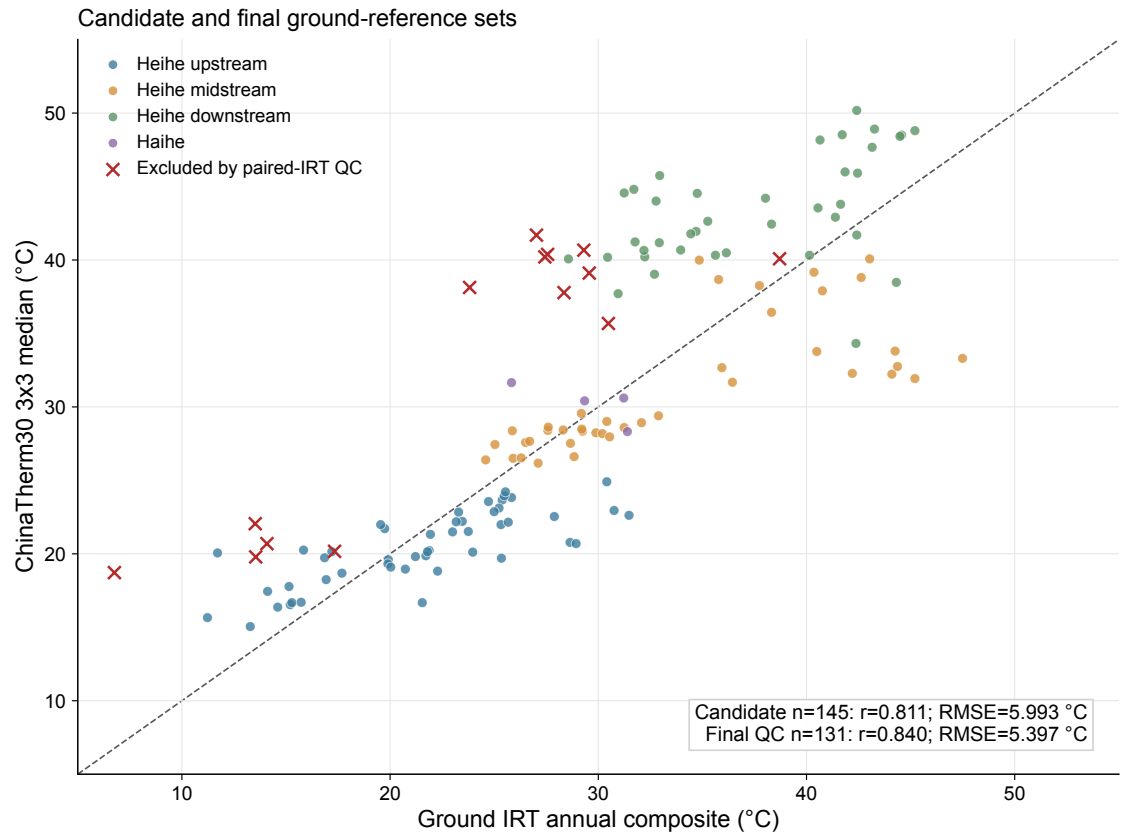


图 S6: ChinaTherm30 3×3 neighbourhood medians compared with annual ground-IRT composites before and after ground-reference quality control. Coloured circles are the 131 final station-years, and red crosses are the 14 records excluded by the paired-IRT criterion. Summary statistics report the complete 145-record candidate set and the 131-record final set.

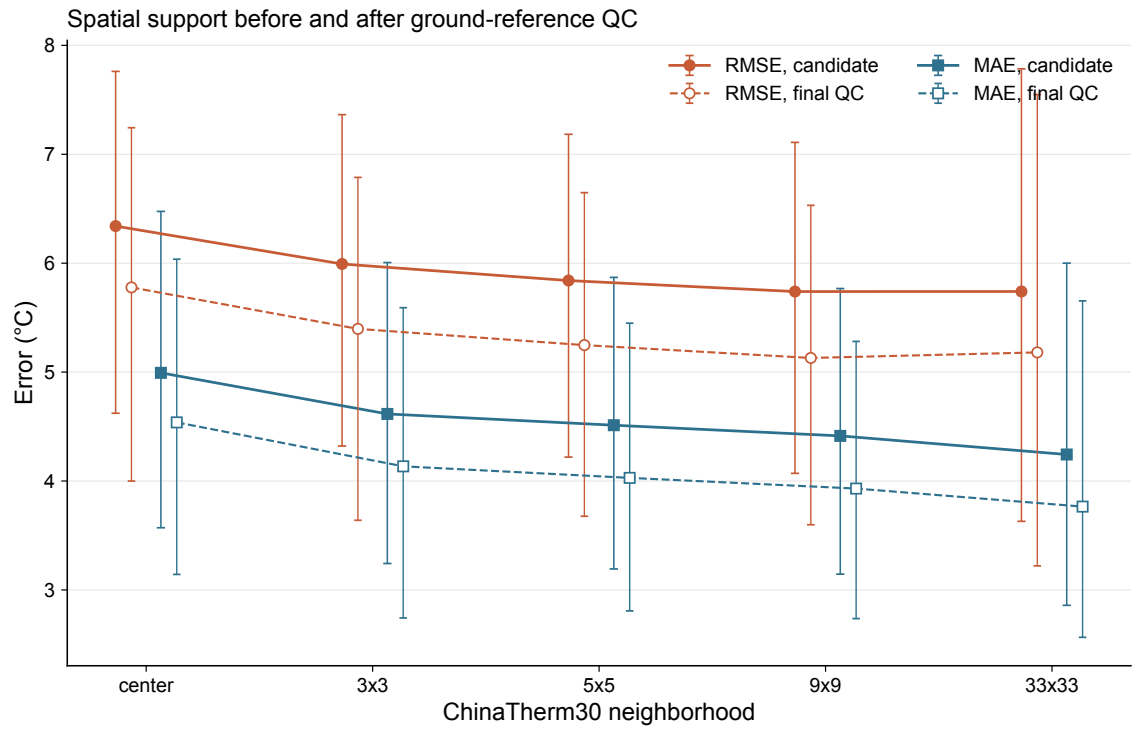


图 S7: Spatial-support sensitivity before and after ground-reference quality control. Points show RMSE and MAE for the ChinaTherm30 centre pixel and median neighbourhoods from 3×3 to 33×33 pixels; error bars are 95% site-cluster bootstrap intervals. The four series are slightly offset horizontally within each neighbourhood for legibility. The main text reports the final quality-controlled set.

Supplementary Tables

表 S1: Station metadata, source parsing, and final validation coverage. Heihe Site IDs 1–23 follow Liu et al. (2018); H1 and H2 denote the Guantao and Daxing sites in the Haihe basin. Candidate records use $K \geq 0.65$, 10:30 local mean solar time ± 30 min, and at least eight valid MOD11A2-aligned eight-day bins; final coverage additionally applies the paired-IRT ground-reference criterion.

Site ID	Station	Network	Reach	Landscape	Lon.	Lat.	Elev.	Duration reported	Parsed XLSX	Final years	Site-years	Status
1	Jingyangling	Heihe	Up.	alpine meadow	101.120	37.840	3750	Aug. 2013-present	13	2014-2024	5	Final QC
2	E'bao	Heihe	Up.	alpine meadow	100.920	37.950	3294	June 2013-Oct. 2016	4	2013-2016	4	Final QC
3	Huangcaogou	Heihe	Up.	alpine meadow	100.730	38.000	3137	June 2013-Apr. 2015	4	2013-2014	2	Final QC
4	Arou Sunny	Heihe	Up.	alpine meadow	100.520	38.090	3529	Aug. 2013-Aug. 2015	3	2013-2015	3	Final QC
5	Arou	Heihe	Up.	subalpine meadow	100.460	38.050	3033	June 2008-present	12	2013-2024	12	Final QC
6	Arou Shady	Heihe	Up.	alpine meadow	100.410	37.980	3536	Aug. 2013-Aug. 2015	3	2013-2014	2	Final QC
7	Yakou	Heihe	Up.	alpine meadow	100.240	38.010	4148	Jan. 2014-present	10	2016-2024	9	Final QC
8	Huangzangsi	Heihe	Up.	wheat	100.190	38.230	2612	June 2013-Apr. 2015	4	2013-2014	2	Final QC
9	Dashalong	Heihe	Up.	alpine marsh	98.940	38.840	3739	Aug. 2013-present	12	2014-2023	10	Final QC
10	Guantan	Heihe	Up.	Qinghai spruce	100.250	38.530	2835	2008-2011	0	–	0	No IRT work-book
11	Huazhaizi	Heihe	Mid.	Kalidium desert	100.320	38.770	1731	June 2012-present	12	2015-2016	2	Final QC
12	Shenshawo	Heihe	Mid.	sandy desert	100.490	38.790	1594	June 2012-Apr. 2015	4	2012-2014	3	Final QC
13	Heihe RS	Heihe	Mid.	grassland	100.480	38.830	1560	Aug. 2014-present	9	2016-2024	9	Final QC
14	Zhangye wetland	Heihe	Mid.	reed	100.450	38.980	1460	June 2012-present	11	2013-2024	11	Final QC
15	Daman	Heihe	Mid.	maize	100.370	38.860	1556	May 2012-present	12	2013-2024	12	Final QC
16	Yingke	Heihe	Mid.	maize	100.410	38.860	1519	2008-2011	0	–	0	No IRT work-book
17	Bajitan Gobi	Heihe	Mid.	Reaumuria	100.300	38.920	1562	May 2012-Apr. 2015	3	2012-2014	3	Final QC
18	Mixed Forest	Heihe	Down.	Populus-Tamarix	101.130	41.990	874	July 2013-present	12	2013-2024	12	Final QC

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Table S1 continued

Site ID	Station	Network	Reach	Landscape	Lon.	Lat.	Elev.	Duration reported	Parsed XLSX	Final years	Site-years	Status
19	Barren Land	Heihe	Down.	bare land	101.130	42.000	878	July 2013-Mar. 2016	2	2015	1	Final QC
20	Sidaoqiao	Heihe	Down.	Tamarix	101.140	42.000	873	July 2013-present	14	2013-2024	12	Final QC
21	Cropland	Heihe	Down.	melon	101.130	42.000	875	July 2013-Nov. 2015	2	2013-2014	2	Final QC
22	Populus	Heihe	Down.	Populus	101.120	41.990	876	July 2013-Apr. 2016	2	2015	1	Final QC
23	Desert	Heihe	Down.	Reaumuria	100.990	42.110	1054	Apr. 2015-present	10	2015-2024	10	Final QC
H1	Guantao	Haihe	–	–	115.117	36.500	–	2008-2010 dataset scope	1	2010	1	Final QC
H2	Daxing	Haihe	–	–	116.416	39.616	–	2008-2010 dataset scope	1	2008-2010	3	Final QC

表 S2: Ground-IRT processing and validation protocol.

Step	Definition
Source screening	Scan 165 XLSX workbooks; retain the 160 files with usable timestamps and IRT variables; preserve source-level audit status, duplicate-file evidence, and station-lineage decisions.
Time basis	Interpret station timestamps as China Standard Time and convert to local mean solar time using station longitude.
Nominal overpass	Use 10:30 local mean solar time with a ± 30 min primary window; ± 60 and ± 90 min are sensitivity tests.
IRT consensus	Average IRT-1 and IRT-2 when both are valid; reject paired observations differing by more than 5 degrees C; retain a single valid sensor when only one is available.
Daily aggregation	Take the median of valid observations within each station-day and overpass window.
Clear-sky protocol	Use downward shortwave radiation to calculate a clear-sky index; $K \geq 0.65$ is primary, with all-sky and $K \geq 0.60$ alternatives.
Eight-day alignment	Assign daily values to MOD11A2 nominal eight-day bins whose start DOY is 150-300; average valid daily values within each bin.
Annual composite	Take the median of qualified eight-day means and require at least eight valid bins per station-year.
Ground-reference QC	Apply the paired-IRT consistency criterion to 145 candidate station-years and retain 131 records across all 23 sites. Exclude a station-year when its median daily paired-IRT absolute difference exceeds 5 degrees C; product values and residuals are not used.
Product extraction	Extract the colocated ChinaTherm30 center pixel and median neighborhoods of 3x3, 5x5, 9x9, and 33x33 pixels.
Metrics and uncertainty	Report Pearson r , RMSE, MAE, bias, site-stratified results, and 2,000-replicate site-cluster bootstrap intervals.

The temporal, clear-sky and annual-support filters yielded 145 candidate station-years. Ground-reference quality control excluded a station-year when the median of its daily paired-IRT absolute differences exceeded 5 °C; no ChinaTherm30 or RF-LST value was used in this decision. The final set retained 131 station-years across all 23 sites and excluded 14 records (eight at Huazhaizi Desert Steppe, five at Jingyangling, and one at Barren Land). For the ChinaTherm30 3×3 comparison, r and RMSE changed from 0.811 and 5.993 °C in the candidate set to 0.840 and 5.397 °C in the final set. Table S3 provides the complete exclusion audit.

表 S3: Station-years excluded from the final ground-reference set by paired-IRT inconsistency. Paired days count days with a finite IRT-1/IRT-2 diagnostic; medians and 95th percentiles are daily absolute paired-sensor differences. No satellite-product residual was used in the criterion.

Station	Chinese name	Year	Paired days	Median (°C)	P95 (°C)	Status
Barren Land	裸地	2013	85	5.190	6.098	Excluded
Huazhaizi	花寨子	2017	112	5.225	7.202	Excluded
Desert Steppe	荒漠					
Huazhaizi	花寨子	2018	108	8.757	11.329	Excluded
Desert Steppe	荒漠					
Huazhaizi	花寨子	2019	117	8.005	13.871	Excluded
Desert Steppe	荒漠					
Huazhaizi	花寨子	2020	122	8.830	14.162	Excluded
Desert Steppe	荒漠					
Huazhaizi	花寨子	2021	109	10.620	15.492	Excluded
Desert Steppe	荒漠					
Huazhaizi	花寨子	2022	89	10.520	13.713	Excluded
Desert Steppe	荒漠					
Huazhaizi	花寨子	2023	120	9.425	13.892	Excluded
Desert Steppe	荒漠					
Huazhaizi	花寨子	2024	115	8.150	12.877	Excluded
Desert Steppe	荒漠					
Jingyangling	景阳岭	2018	107	6.305	10.326	Excluded
Jingyangling	景阳岭	2019	115	7.570	16.906	Excluded
Jingyangling	景阳岭	2020	110	8.273	12.861	Excluded
Jingyangling	景阳岭	2022	106	5.447	9.921	Excluded
Jingyangling	景阳岭	2023	121	6.960	9.720	Excluded

References

- Liu, S., Li, X., Xu, Z., Che, T., Xiao, Q., Ma, M., Liu, Q., Jin, R., Guo, J., Wang, L., Wang, W., Qi, Y., Li, H., Xu, T., Ran, Y., Hu, X., Shi, S., Zhu, Z., Tan, J., ... Ren, Z. (2018). The Heihe Integrated Observatory Network: A basin-scale land surface processes observatory in China. *Vadose Zone Journal*, 17(1), 180072. <https://doi.org/10.2136/vzj2018.04.0072>